

AI for Cybersecurity Operations & Incident Response

Production Assignments

DAY 1 Assignment

SOC Log Investigation with Python

Objective

Participants act as **L1 SOC analysts** parsing messy logs to detect suspicious activity.

Assignment 1 — Firewall Log Investigation

Input (sample log file)

```
2025-01-10 10:21:02 ACCEPT src=192.168.1.15 dst=10.0.0.8 port=443 user=admin
2025-01-10 10:21:05 DROP src=45.33.21.9 dst=10.0.0.8 port=22 user=root
2025-01-10 10:21:07 DROP src=45.33.21.9 dst=10.0.0.8 port=22 user=root
2025-01-10 10:21:09 DROP src=45.33.21.9 dst=10.0.0.8 port=22 user=root
2025-01-10 10:22:10 ACCEPT src=192.168.1.20 dst=10.0.0.8 port=443 user=venkat
```

Tasks

Students must write a Python script that:

Extracts

- timestamp
- source IP

- destination IP
 - port
 - user
 - action
-

② Detect suspicious activity

Conditions

> More than 3 DROP events from same IP

③ Convert logs to structured JSON

Expected output

```
{  
  "timestamp": "2025-01-10 10:21:05",  
  "src_ip": "45.33.21.9",  
  "dest_ip": "10.0.0.8",  
  "port": 22,  
  "user": "root",  
  "action": "DROP"  
}
```

Deliverables

Participants must submit

log_parser.py
normalized_logs.json
suspicious_ips.txt

DAY 2 Assignment

Detection Engineering Challenge

Objective

Participants act as **SOC detection engineers** reducing false positives.

Assignment 2 — Build Detection Rules

Input dataset

```
timestamp,user,login_status,geo,device
10:01,admin,FAIL,Russia,Unknown
10:02,admin,FAIL,Russia,Unknown
10:03,admin,FAIL,Russia,Unknown
10:05,admin,SUCCESS,India,Laptop
10:07,venkat,SUCCESS,India,Mobile
```

Tasks

Build Python logic detecting

Rule 1

Multiple failed logins (>3 within 5 minutes)

Rule 2

Impossible travel

Login from Russia then India within 10 minutes

Rule 3

New device access for privileged user

Expected Output

ALERT 1: Brute Force Attempt

User: admin

IP: Russia

ALERT 2: Impossible Travel

User: admin

Russia -> India

DAY 2 Assignment (Prompt Engineering)

Assignment 3 — SOC Prompt Engineering

Participants must design **3 prompts**.

Prompt 1 — Incident Classification

Input

User admin failed login 15 times from Russia then logged in successfully

Expected LLM output

```
{  
  "incident_type": "Credential Brute Force",  
  "severity": "High",  
  "confidence": 0.92  
}
```

Prompt 2 — False Positive Reduction

Input

User login from US then India within 1 hour

Expected Output

```
{  
  "classification": "Likely False Positive",  
  "reason": "VPN usage possible"  
}
```

Prompt 3 — SOC Triage

Input

Multiple API failures from merchant gateway

Expected Output

```
{  
  "incident_type": "API Abuse",  
  "recommended_action": "Rate limit API key"  
}
```

Deliverable

soc_prompts.md

DAY 3 Assignment

ML Anomaly Detection for Payments

Objective

Build ML model detecting abnormal transaction behavior.

Dataset

transaction_id,user,amount,geo,time

T1,A,50,India,10

T2,A,60,India,12

T3,A,55,India,15

T4,A,5000,Russia,16

Tasks

Participants must:

1 Build Isolation Forest model

Detect anomalous transactions.

2 Generate risk score

Example

Transaction T4

Risk Score: 0.93

Reason: Amount deviation + Geo anomaly

3 Output suspicious transactions

ALERT: Possible Fraud Transaction

User: A

Amount: 5000

Geo: Russia

Deliverables

anomaly_detection.py

risk_scores.csv

DAY 4 Assignment

Incident Response Automation

Objective

Automate IR documentation.

Scenario

API gateway receiving abnormal traffic
Login failures spike
Multiple merchant accounts affected

Tasks

Participants must build Python script that:

1 Generates IR timeline

Example

10:01 API traffic spike detected
10:03 Login failures increase
10:05 Possible credential stuffing

2 Suggest containment actions

Rate limit API
Block suspicious IP
Force password reset

③ Generate IR report

Output format

incident_report.txt

Structure

Incident Summary

Impact

Timeline

Containment Actions

Recommendations

DAY 5 Capstone Assignment

FISERV-Style SOC Investigation

Participants must solve a **realistic incident scenario**.

Scenario

Payment system shows:

API spike

Failed logins

Declined transactions

Participants Must Build

Pipeline

Logs → Parsing → Feature Engineering → ML → LLM Analysis → IR Report

Tasks

① Parse logs

payment_logs.txt
api_logs.txt
auth_logs.txt

② Mask sensitive data

4111-XXXX-XXXX-1234

③ Detect anomalies

Transaction spikes
API abuse
Credential stuffing

④ Use LLM prompt to classify

Cyber Attack
Fraud
Operational issue

⑤ Generate SOC Report

Example

Incident Type: Credential Stuffing Attack

Impact:

12 merchant accounts affected

Recommended Action:
Block IP range
Force password reset
Enable MFA

Final Deliverables

Participants must submit

project/

log_parser.py
anomaly_model.py
prompt_engine.py
incident_report_generator.py
soc_final_report.pdf

Evaluation Rubric

Criteria	Weight
Log Parsing Accuracy	20%
Detection Logic	20%
ML Model	20%
Prompt Engineering	20%
IR Documentation	20%